

COMPOUND INTEREST (Stage 1)

[Basic Concepts]

2.1 INTRODUCTION

Sometimes, in need, we borrow money from a bank or some other agency doing financial business. In general, the money is borrowed for a specified period and has to be returned at the end of that period. At the end of the period, we pay the money borrowed **plus** some extra money for utilising the money of the lender.

The money borrowed is called the **principal**, the extra money paid for using lender's money is called **interest** and the total money, paid to the lender at the end of the specified period is called the **amount**.

$$\text{Amount} = \text{Principal} + \text{Interest i.e. } A = P + I$$

2.2 INTEREST (Simple Interest)

Interest is said to be simple, if it is calculated on the original principal throughout the loan period, irrespective of the length of the period for which it is borrowed.

1. We know, $S. I. = \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$ i.e. $I = \frac{P \times R \times T}{100}$
2. When we say, **interest**, it always means **simple interest**.
3. On the same sum and at the same rate of interest per annum, the interest of every year is same.

2.3 COMPOUND INTEREST (C.I.)

Money is said to be lent at **compound interest**, when the interest, which has become due at the end of a certain fixed period (one year, half year, etc., as given), is not paid to the money lender, but is added to the sum lent. The amount thus obtained becomes the principal for the next period. This process is repeated until the amount for the last period is found.

The difference between the final amount and the original principal is the required **compound interest**.

$$\therefore \text{Compound Interest} = \text{Final Amount} - \text{Original Principal i.e. } C.I. = A - P$$

The difference between **simple interest** (S.I.) and **compound interest** (C.I.), is made clear by the table given on next page.

[Here, in the table, we have taken sum borrowed (principal)] = ₹ 1,000 at 10% per annum and for 3 years.]

At simple interest		At compound interest	
<u>For 1st year</u>	$P = ₹ 1,000$ $I = \frac{₹ 1,000 \times 10 \times 1}{100}$ $= ₹ 100 \text{ (S.I.)}$ $\therefore \text{Amount} = ₹ 1,000 + ₹ 100$ $= ₹ 1,100$		$P = ₹ 1,000$ $I = \frac{₹ 1,000 \times 10 \times 1}{100}$ $= ₹ 100 \text{ (C.I.)}$ $\therefore \text{Amount} = ₹ 1,000 + ₹ 100$ $= ₹ 1,100$
		For 1st year : C.I. = S.I.	
<u>For 2nd year</u>	$P = ₹ 1,000$ $I = \frac{₹ 1,000 \times 10 \times 1}{100}$ $= ₹ 100 \text{ (S.I.)}$ $\therefore \text{Amount} = ₹ 1,100 + ₹ 100$ $= ₹ 1,200$		$P = ₹ 1,100$ $I = \frac{₹ 1,100 \times 10 \times 1}{100}$ $= ₹ 110 \text{ (C.I.)}$ $\therefore \text{Amount} = ₹ 1,100 + ₹ 110$ $= ₹ 1,210$
		For 2nd year : C.I. is more than the S.I.	
<u>For 3rd year</u>	$P = ₹ 1,000$ $I = \frac{₹ 1,000 \times 10 \times 1}{100}$ $= ₹ 100 \text{ (S.I.)}$ $\therefore \text{Amount} = ₹ 1,200 + ₹ 100$ $= ₹ 1,300$		$P = ₹ 1,210$ $I = \frac{₹ 1,210 \times 10 \times 1}{100}$ $= ₹ 121 \text{ (C.I.)}$ $\therefore \text{Amount} = ₹ 1,210 + ₹ 121$ $= ₹ 1,331$
Every year, C.I. increases but the S.I. remains the same.			

2.4

COMPOUND INTEREST AS A REPEATED SIMPLE INTEREST COMPUTATION WITH A GROWING PRINCIPAL

As shown in the table, given above, the principal for 1st year is ₹ 1,000 and interest (C.I.) on it is ₹ 100. The principal for 2nd year is ₹ 1,100 and interest (C.I.) on it is ₹ 110; whereas, the principal for 3rd year is ₹ 1,210 and the interest (C.I.) on it is ₹ 121.

It is observed that the compound interest is growing (increasing) every year which increases the principal for next year.

As shown in the table, given above, **compound interest in 3 years**

$$= \text{C.I. of 1st year} + \text{C.I. of 2nd year} + \text{C.I. of 3rd year}$$

$$= ₹ 100 + ₹ 110 + ₹ 121 = ₹ 331$$

Also, compound interest in 3 years.

$$= \text{Amount at the end of 3 years} - \text{Original sum (Principal for 1st year)}$$

$$= ₹ 1,331 - ₹ 1,000 = ₹ 331$$

1 ₹ 8,000 is lent at 5 percent compound interest per year for 2 years. Find the amount and the compound interest.

Solution :

For the first year :

$$\begin{aligned} \text{Principal (P)} &= ₹ 8,000; \text{Rate (R)} = 5\% \\ \text{and, Time (T)} &= 1 \text{ year} \\ \therefore \text{Interest} &= \frac{P \times R \times T}{100} \\ &= \frac{₹ 8,000 \times 5 \times 1}{100} = ₹ 400 \\ \text{Amount} &= \text{Principal} + \text{Interest} \\ &= ₹ 8,000 + ₹ 400 = ₹ 8,400 \end{aligned}$$

According to the definition of compound interest, the amount of the first year will work as principal for the next (second) year.

∴ For the second year :

$$\text{Principal (P)} = ₹ 8,400; R = 5\% \text{ and } T = 1 \text{ year}$$

$$\text{Interest} = ₹ \frac{8,400 \times 5 \times 1}{100} = ₹ 420$$

$$\therefore I = \frac{P \times R \times T}{100}$$

$$\text{Amount at the end of 2nd year} = ₹ 8,400 + ₹ 420 = ₹ 8,820$$

Ans.

$$\begin{aligned} \text{Compound Interest} &= \text{Final Amount} - \text{Initial Principal} \\ &= ₹ 8,820 - ₹ 8,000 = ₹ 820 \end{aligned}$$

Ans.

Also, **C.I. of 2 years** = C.I. of 1st year + C.I. of 2nd year
= ₹ 400 + ₹ 420 = ₹ 820

Ans.

Using formula :

When the interest is compounded yearly, the formula for finding the amount is :

$$A = P \left(1 + \frac{r}{100} \right)^n$$

where A = amount; P = principal;
r = rate of interest compounded yearly;
and n = number of years.

For example 1, given above, we have P = ₹ 8,000, r = 5% and n = 2 years

$$\therefore \text{Amount (A)} = P \left(1 + \frac{r}{100} \right)^n$$

$$\begin{aligned} \Rightarrow A &= ₹ 8,000 \left(1 + \frac{5}{100} \right)^2 \\ &= ₹ 8,000 \times \left(\frac{21}{20} \right)^2 \\ &= ₹ 8,000 \times \frac{441}{400} = ₹ 8,820 \end{aligned}$$

Ans.

$$\begin{aligned} \text{C.I.} &= \text{Amount} - \text{Principal} \\ &= ₹ 8,820 - ₹ 8,000 = ₹ 820 \end{aligned}$$

Ans.

2 Find the amount and the compound interest on ₹ 10,000 at 8 per cent per annum and in year; interest being compounded half-yearly.

For the last $\frac{1}{2}$ year : $P = ₹ 7,260$, $R = 10\%$ and $T = \frac{1}{2}$ year

$$\therefore I = ₹ \frac{7,260 \times 10 \times 1}{100 \times 2} = ₹ 363$$

$$\text{and amount, } A = ₹ 7,260 + ₹ 363 = ₹ 7,623$$

Since, the amount in $2\frac{1}{2}$ years is ₹ 7,623 and the original principal is ₹ 6,000

$$\therefore \text{Compound Interest} = ₹ 7,623 - ₹ 6,000 = ₹ 1,623$$

Using formula :

In the example 3, given above

First of all find the amount in 2 years.

$$\text{Amount in 2 years} = ₹ 6,000 \left(1 + \frac{10}{100}\right)^2 = ₹ 7,260$$

After two years, ₹ 7,260 is the principal for the remaining half year and so :

$$\begin{aligned} A &= P \left(1 + \frac{r}{2 \times 100}\right)^{n \times 2} \\ &= ₹ 7,260 \left(1 + \frac{10}{2 \times 100}\right)^{\frac{1}{2} \times 2} = ₹ 7,623 \end{aligned}$$

$$\Rightarrow \text{Amount in } 2\frac{1}{2} \text{ years} = ₹ 7,623$$

$$\therefore \text{C.I.} = ₹ 7,623 - ₹ 6,000 = ₹ 1,623$$

Alternative method :

For the example given above, we can directly use the formula :

$$A = P \left(1 + \frac{r}{100}\right)^2 \left(1 + \frac{r}{2 \times 100}\right)^{\frac{1}{2} \times 2}$$

$$\Rightarrow \text{Amount in } 2\frac{1}{2} \text{ years} = ₹ 6,000 \left(1 + \frac{10}{100}\right)^2 \left(1 + \frac{10}{2 \times 100}\right) = ₹ 7,623$$

Study the following table carefully :

No. of years :	Compounded yearly :	Compounded half-yearly :
(i) $n = 1$ year	$A = P \left(1 + \frac{r}{100}\right)^1$	$A = P \left(1 + \frac{r}{2 \times 100}\right)^{1 \times 2}$
(ii) $n = 1\frac{1}{2}$ years	$A = P \left(1 + \frac{r}{100}\right)^1 \cdot \left(1 + \frac{r}{2 \times 100}\right)^{\frac{1}{2} \times 2}$	$A = P \left(1 + \frac{r}{2 \times 100}\right)^{\frac{3}{2} \times 2}$
(iii) $n = 2$ years	$A = P \left(1 + \frac{r}{100}\right)^2$	$A = P \left(1 + \frac{r}{2 \times 100}\right)^{2 \times 2}$
iv) $n = 2\frac{1}{2}$ years	$A = P \left(1 + \frac{r}{100}\right)^2 \cdot \left(1 + \frac{r}{2 \times 100}\right)^{\frac{1}{2} \times 2}$	$A = P \left(1 + \frac{r}{2 \times 100}\right)^{\frac{5}{2} \times 2}$

4 Calculate the compound interest accrued on ₹ 16,000 in 3 years, when the rates of interest for successive years are 10%, 12% and 15% respectively.

Amount -
Principal for 2nd year -

$$\text{For 2nd year } P = ₹ 4,320$$

$$\text{Interest on it} = ₹ \frac{4,320 \times 8 \times 1}{100} = ₹ 345.60$$

$$\text{Amount of 2nd year} = ₹ 4,320 + ₹ 345.60 \\ = ₹ 4665.60$$

$$\text{C.I. of 2 years} = ₹ 4,665.60 - ₹ 4,000 = ₹ 665.60$$

$$\text{Required difference between C.I. and S.I.} = \text{C.I.} - \text{S.I.} \\ = ₹ 665.60 - ₹ 640 = ₹ 25.60$$

Ans.

Alternative method :

In the example 5, given above

$$\text{S.I.} = ₹ \frac{4,000 \times 8 \times 2}{100} = ₹ 640$$

$$A = ₹ 4,000 \times \left(1 + \frac{8}{100}\right)^2$$

$$= ₹ 4,000 \times \frac{27}{25} \times \frac{27}{25} = ₹ 4,665.60$$

$$\text{C.I.} = \text{Amount} - \text{Principal}$$

$$= ₹ 4,665.60 - ₹ 4,000 = ₹ 665.60$$

Ans.

Required difference between C.I. and S.I.

$$= \text{C.I.} - \text{S.I.}$$

$$= ₹ 665.60 - ₹ 640 = ₹ 25.60$$

Ans.

It is clear from examples, given above, that :

1. when the interest is compounded yearly, the principal changes (increases) every year.
2. when the interest is compounded half-yearly, the principal increases every six months.
3. The period (time), after which the principal changes, is called the **conversion period**.

In example 1, given above, the **conversion period** is **one year**.

And, in example 2, given above, the **conversion period** is **half-year**.

EXERCISE 2 (A)

1. Multiple Choice Type :

Choose the correct answer from the options given below.

- (a) For a particular year the simple interest at 10% is ₹ 800. The compound interest for the next year at the same rate is :
- (i) ₹ 880 (ii) ₹ 800
(iii) ₹ 720 (iv) ₹ 968

- (b) The compound interest on ₹ 5,000 at 10% per annum and in 6 months amounts to :

- (i) ₹ 5,500 (ii) ₹ 250
(iii) ₹ 600 (iv) ₹ 2,500

- (c) The compound interest on ₹ 5,000 at 10% per annum and in one year compounded half-yearly is :

- (i) ₹ 1,050 (ii) ₹ 525
(iii) ₹ 512.50 (iv) ₹ 5,512.50

(d) A sum of ₹ 20,000 is lent at 12% compound interest compounded yearly. The compound interest accrued in the second year will be :

- (i) ₹ 4,800 (ii) ₹ 288
(iii) ₹ 2,688 (iv) ₹ 5,088

(e) During the year 2022, the interest accrued at the rate of 5% is ₹ 1,250. The compound interest accrued at the same rate during the year 2023 is :

- (i) ₹ 1,312.50 (ii) ₹ 62.50
(iii) ₹ 6,250 (iv) ₹ 2,000

(f) Rates of interest for two consecutive years are 10% and 12% respectively. The percentage increase during these two years is :

- (i) 22% (ii) 23.2%
(iii) 123.2% (iv) 122%

2. ₹ 16,000 is invested at 5% compound interest compounded per annum.

Use the table, given below, to find the amount in 4 years.

Year↓	Initial amount (₹)	Interest (₹)	Final amount (₹)
1st	...16,000...	8,00.....	...16,800...
2nd			
3rd			
4th			
5th			

3. Calculate the amount and the compound interest on ₹ 8,000 in $2\frac{1}{2}$ years at 15% per annum.

4. Calculate the amount and the compound interest on :

₹ 4,600 in 2 years when the rates of interest of successive years are 10% and 12% respectively.

5. Meenal lends ₹ 75,000 at C.I. for 3 years. If the rate of interest for the first two years is 15% per year and for the third year it is 16%, calculate the sum Meenal will get at the end of the third year.

6. Calculate the amount and the compound interest on ₹ 16,000 in 3 years, when the rates of the interest for successive years are 10%, 14% and 15% respectively.

7. Calculate the compound interest for the second year on ₹ 8,000/- invested for 3 years at 10% per annum.

8. Find the compound interest, correct to the nearest rupee, on ₹ 2,400 for $2\frac{1}{2}$ years at 5 percent per annum.

9. A borrowed ₹ 2,500 from B at 12% per annum compound interest. After 2 years, A gave ₹ 2,936 and a watch to B to clear the account. Find the cost of the watch. (LS)

10. How much will ₹ 50,000 amount to in 3 years, compounded yearly, if the rates for the successive years are 6%, 8% and 10% respectively.

11. Govind borrows ₹ 18,000 at 10% simple interest. He immediately invests the money borrowed at 10% compound interest compounded half-yearly. How much money does Govind gain in one year ? (HOTS)

12. Find the compound interest on ₹ 4,000 accrued in three years, when the rate of interest is 8% for the first year and 10% per year for the second and the third years.

6 Mrs. Kapoor invested ₹ 6,000 every year at the beginning of the year, at 10% per annum compound interest. Calculate the amount of her total savings :

- (i) upto the end of the second year. (ii) at the beginning of the third year.

Solution :

(i) **For 1st year**

Since, money invested at the beginning of the year = ₹ 6,000

₹ 6,000

And, amount of 1st year = ₹ 100 + ₹ 10 = ₹ 110

⇒ The principal for 2nd year = ₹ 110

C.I. of 2nd year = ₹ $\frac{110 \times 10 \times 1}{100}$ = ₹ 11

And, amount of 2nd year = ₹ (110 + 11) = ₹ 121

⇒ The principal for 3rd year = ₹ 121

C.I. of 3rd year = ₹ $\frac{121 \times 10 \times 1}{100}$ = ₹ 12.10

Sum of the C.I. of 1st year and C.I. of 3rd year

= ₹ 10 + ₹ 12.10 = ₹ 22.10

Now, when sum of two interests = ₹ 22.10, principal (sum) = ₹ 100

And, when sum of two interests = ₹ 1,768, sum = ₹ $\frac{100}{22.10} \times 1,768$ = ₹ 8,000 Ans.

EXERCISE 2 (B)

1. Multiple Choice Type :

Choose the correct answer from the options given below.

- (a) The difference between C.I. and S.I. in one year on ₹ 5,000 at the rate of 10% per annum is :
 - (i) ₹ 00
 - (ii) ₹ 500
 - (iii) ₹ 5,500
 - (iv) ₹ 5,250
 - (b) ₹ 2,000 is saved during the year 2022 and deposited in a bank at the beginning of year 2023 at 8% compound interest. During 2023, ₹ 3,000 more is saved and deposited in the same bank at the beginning of 2024 and at the same rate of interest. The C.I. earned upto the end of 2024 is :
 - (i) ₹ 560
 - (ii) ₹ 572.80
 - (iii) ₹ 5,400
 - (iv) ₹ 400
 - (c) ₹ 1,000 is borrowed at 10% per annum C.I. If ₹ 300 is repaid at the end of each year, the amount of loan outstanding at the end of 2nd year is :
 - (i) ₹ 880
 - (ii) ₹ 610
 - (iii) ₹ 580
 - (iv) ₹ 484
 - (d) The difference between C.I. and S.I. in 2 years on ₹ 4,000 at 10% per annum is :
 - (i) ₹ 840
 - (ii) ₹ 800
 - (iii) ₹ 400
 - (iv) ₹ 40
 - (e) ₹ 10 is the difference between compound interest and simple interest in 2 years and at 5% per annum. The principal amount is :
 - (i) ₹ 4,400
 - (ii) ₹ 4,100
 - (iii) ₹ 4,000
 - (iv) none of these
2. Calculate the difference between the simple interest and the compound interest on ₹ 4,000 in 2 years at 8% per annum compounded yearly.
 3. A sum of money is lent at 8% per annum compound interest. If the interest for the second year exceeds that for the first year by ₹ 96, find the sum of money.
 4. A man invests ₹ 5,600 at 14% per annum compound interest for 2 years. Calculate :
 - (i) the interest for the first year.
 - (ii) the amount at the end of the first year.
 - (iii) the interest for the second year, correct to the nearest rupee.
 5. A man saves ₹ 3,000 every year and invests it at the end of the year at 10% compound interest. Calculate the total amount of his savings at the end of the third year.
 6. A man lends ₹ 12,500 at 12% for the first year, at 15% for the second year and at 18% for the third year. If the rates of interest are compounded yearly; find the difference between the C.I. of the first year and the compound interest for the third year.
 7. A man borrows ₹ 6,000 at 5 percent C.I. per annum. If he repays ₹ 1,200 at the end of each year, find the amount of the loan outstanding at the beginning of the third year. (LS)
 8. A man borrows ₹ 5,000 at 12 percent compound interest payable every six months. He repays ₹ 1,800 at the end of every six